



The Parametric Complexity of Operator Learning

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1. Parametric PDEs and FEONets

Equation: Given $a(x)$, $b(x)$, $c(x)$, input parameter $f(x)$, output solution u :

$$-\operatorname{div}(a(x)\nabla u) + b(x) \cdot \nabla u + c(x)u = f(x) \quad \text{in } D, \quad u|_{\partial D} = g(x). \quad (1)$$

Objective: Estimate the error between the true solution and the FEONets solution.

1. Supervised learning (DeepONet, FNO): $\|u_{\text{NN}}(\mathbf{f}; \theta) - u_{\text{trad}}\|^2$

u_{trad} is the solution obtained by traditional numerical methods, e.g., FEM $Au_{\text{trad}} = L$

$$\|u_{\text{NN}}(\mathbf{f}; \theta) - A^{-1}L\|^2 \quad (2)$$

2. Unsupervised learning (FEONet): $\|Au_{\text{NN}}(\mathbf{f}; \theta) - L\|^2$



1. FEONets

$$f(\mathbf{x}, \omega) = \omega_1 \sin 2\pi\omega_2\mathbf{x} + \omega_3 \cos 2\pi\omega_4\mathbf{x}$$

Input: $\omega = (\omega_1, \omega_2, \omega_3, \omega_4)$, i.i.d. uniform distribution on $\Omega := [0, 1]^4$, (**More general input:** $\{f(\mathbf{x}_i)\}_{i=1}^N$)

Solution structure: $u_{h,\theta}(\mathbf{x}, \omega) = \sum_{k=1}^{N_h} \alpha_k(\omega) \phi_k(\mathbf{x})$, $\{\phi_k\}_{k=1}^{N_h}$ is the finite element basis

Output: coefficients $\alpha(\omega) := (\alpha_1(\omega), \alpha_2(\omega), \dots, \alpha_{N_h}(\omega))$

Loss function: $\mathcal{L}(\alpha) := \int_{\Omega} \|A\alpha(\omega) - L\|_{\ell^2} dP(\omega)$, A is the stiffness matrix, L is the load vector.

Monte Carlo approximation: $\mathcal{L}^M(\alpha) = \frac{|\Omega|}{M} \sum_{j=1}^M \|A\alpha(\omega^j) - L\|_{\ell^2}$, $\{\omega^j\}_{j=1}^M$ are i.i.d. samples from Ω



1. Neural network architecture

Neural network architecture: For each $L \in \mathbb{N}$, an L -layer ReLU neural network:

$f^L(x) : \mathbb{R}^{n_0} \rightarrow \mathbb{R}^{n_L}$ is defined by

$$f^1(x) = W^1x + b^1, \quad f^\ell(x) = W^\ell \sigma(f^{\ell-1}(x)) + b^\ell, \quad \ell = 2, 3, \dots, L, \quad (3)$$

where $W^\ell \in \mathbb{R}^{n_\ell \times n_{\ell-1}}$, $b^\ell \in \mathbb{R}^{n_\ell}$, and $\sigma(x) = \max\{0, x\}$ is the ReLU activation function.

Universal approximation theorem: Suppose there is a sequence of NN spaces $\{\mathcal{N}_n\}_{n=1}^\infty$ such that $\mathcal{N}_n \subset \mathcal{N}_{n+1}$. Let K be a compact set in $\mathbb{R}^m$ and $g \in \mathcal{C}(K, \mathbb{R}^N)$. Then there holds

$$\lim_{n \rightarrow \infty} \inf_{\hat{g} \in \mathcal{N}_n} \|\hat{g} - g\|_{\mathcal{C}(K)} = 0. \quad (4)$$



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2. Analytic framework for convergence

Exact FEM solution: $u_h(\mathbf{x}, \omega) = \sum_{k=1}^{N_h} \alpha_k^*(\omega) \phi_k(\mathbf{x})$, i.e., $A\alpha^*(\omega) = L$.

approximation without quadrature:

$$\alpha_n^{\mathcal{L}} := \arg \min_{\alpha \in \mathcal{N}_n} \mathcal{L}(\alpha) = \arg \min_{\alpha \in \mathcal{N}_n} \int_{\Omega} \|A\alpha(\omega) - L\|_{\ell^2} dP(\omega)$$
$$u_{h,n}(\mathbf{x}, \omega) = \sum_{k=1}^{N_h} (\alpha_n^{\mathcal{L}})_k(\omega) \phi_k(\mathbf{x}). \quad (5)$$

Final approximation:

$$\alpha_{n,M}^{\mathcal{L}} := \arg \min_{\alpha \in \mathcal{N}_n} \mathcal{L}^M(\alpha) = \arg \min_{\alpha \in \mathcal{N}_n} \frac{|\Omega|}{M} \sum_{j=1}^M \|A\alpha(\omega^j) - L\|_{\ell^2}$$
$$u_{h,n,M}(\mathbf{x}, \omega) = \sum_{k=1}^{N_h} (\alpha_{n,M}^{\mathcal{L}})_k(\omega) \phi_k(\mathbf{x}). \quad (6)$$



2. Analytic framework for convergence

$$\alpha^* \sim \mathbf{u}_h, \quad \alpha_n^{\mathcal{L}} \sim \mathbf{u}_{h,n}, \quad \alpha_{n,M}^{\mathcal{L}} \sim \mathbf{u}_{h,n,M}.$$

Error decomposition: $\mathbf{u} - \mathbf{u}_{h,n,M} = (\mathbf{u} - \mathbf{u}_h) + (\mathbf{u}_h - \mathbf{u}_{h,n}) + (\mathbf{u}_{h,n} - \mathbf{u}_{h,n,M}).$

$$\begin{aligned} \|\mathbf{u}_h - \mathbf{u}_{h,n,M}\|_{L^1(\Omega; L^2(D))} &= \int_{\Omega} \left\| \sum_{j=1}^{N_h} (\alpha_j^* - (\alpha_{n,M}^{\mathcal{L}})_j) \phi_j \right\|_{L^2(D)} dP(\omega) \\ &\leq \int_{\Omega} \sum_{j=1}^{N_h} |\alpha_j^* - (\hat{\alpha}_{n,M}^{\mathcal{L}})_j| \|\phi_j\|_{L^2(D)} dP(\omega) \\ &\leq \mathcal{C} N_h^{\frac{1}{2}} \|\alpha^* - \alpha_{n,M}^{\mathcal{L}}\|_{L^1(\Omega; \ell^2)} \max_{1 \leq j \leq N_h} \|\phi_j\|_{L^2(D)} \\ &\leq \mathcal{C} \|\alpha^* - \alpha_{n,M}^{\mathcal{L}}\|_{L^1(\Omega; \ell^2)} \\ &\leq \mathcal{C} (\|\alpha^* - \alpha_n^{\mathcal{L}}\|_{L^1(\Omega; \ell^2)} + \|\alpha_n^{\mathcal{L}} - \alpha_{n,M}^{\mathcal{L}}\|_{L^1(\Omega; \ell^2)}). \end{aligned} \tag{7}$$



2. Estimate of $\|\alpha^* - \alpha_n^{\mathcal{L}}\|_{L^1(\Omega; \ell^2)}$

$\rho_{\min}, \rho_{\max}$: the minimum and maximum singular values of A .

$$\begin{aligned}\|\alpha^* - \alpha_n^{\mathcal{L}}\|_{L^1(\Omega; \ell^2)} &\leq \frac{1}{\rho_{\min}} \|A\alpha^* - A\alpha_n^{\mathcal{L}}\|_{L^1(\Omega; \ell^2)} = \frac{1}{\rho_{\min}} \|A\alpha_n^{\mathcal{L}} - L\|_{L^1(\Omega; \ell^2)} = \frac{1}{\rho_{\min}} \mathcal{L}(\alpha_n^{\mathcal{L}}) \\ &= \frac{1}{\rho_{\min}} \inf_{\alpha \in \mathcal{N}_n} \|A\alpha - L\|_{L^1(\Omega; \ell^2)} \leq \frac{\rho_{\max}}{\rho_{\min}} \inf_{\alpha \in \mathcal{N}_n} \|\alpha - \alpha^*\|_{L^1(\Omega; \ell^2)} \\ &= \kappa(A) \inf_{\alpha \in \mathcal{N}_n} \|\alpha - \alpha^*\|_{L^1(\Omega; \ell^2)} \rightarrow 0 \quad \text{as } n \rightarrow \infty,\end{aligned}\tag{8}$$

where $\kappa(A) = \frac{\rho_{\max}}{\rho_{\min}}$ is the condition number of A .



2. Estimate of $\|\alpha_n^{\mathcal{L}} - \alpha_{n,M}^{\mathcal{L}}\|_{L^1(\Omega;\ell^2)}$

Rademacher complexity: For a function class $\mathcal{F}$ and a sample $S = \{X_1, X_2, \dots, X_M\}$, the empirical Rademacher complexity is defined as

$$\mathcal{R}_M(\mathcal{F}) = \mathbb{E}_{\{X_i\}_{i=1}^M} \mathbb{E}_{\{\varepsilon_i\}_{i=1}^M} \left[\sup_{f \in \mathcal{F}} \frac{1}{M} \sum_{i=1}^M \varepsilon_i f(X_i) \right], \quad (9)$$

where ε_i are i.i.d. Rademacher random variables, $\mathbb{P}(\varepsilon_i = 1) = \mathbb{P}(\varepsilon_i = -1) = \frac{1}{2}$.

$$\mathbb{E}_{\{X_i\}_{i=1}^M} \left[\sup_{f \in \mathcal{F}} \left| \frac{1}{M} \sum_{i=1}^M f(X_i) - \mathbb{E}_{X \sim \mathbb{P}_\Omega} [f(X)] \right| \right] \leq 2\mathcal{R}_M(\mathcal{F}). \quad (10)$$

Estimate: $\mathbb{E}[|\mathcal{L}^M(\alpha) - \mathcal{L}(\alpha)|] \leq 2\mathcal{R}_M(\mathcal{F}_n^{\mathcal{L}})$, where $\mathcal{F}_n^{\mathcal{L}} = \{\|A\alpha - L\|_{\ell^2} : \alpha \in \mathcal{N}_n^*\}$.



2. Estimate of $\|\alpha_n^{\mathcal{L}} - \alpha_{n,M}^{\mathcal{L}}\|_{L^1(\Omega;\ell^2)}$ (continued)

$$\begin{aligned}
 \mathbb{E}[\|\alpha_n^{\mathcal{L}} - \alpha_{n,M}^{\mathcal{L}}\|_{L^1(\Omega;\ell^2)}] &\leq \frac{1}{\rho_{\min}} \mathbb{E}[\|A\alpha_n^{\mathcal{L}} - A\alpha_{n,M}^{\mathcal{L}}\|_{L^1(\Omega;\ell^2)}] \\
 &\leq \frac{1}{\rho_{\min}} \mathbb{E}[\|A\alpha_n^{\mathcal{L}} - L\|_{L^1(\Omega;\ell^2)} + \|A\alpha_{n,M}^{\mathcal{L}} - L\|_{L^1(\Omega;\ell^2)}] \\
 &= \frac{1}{\rho_{\min}} \mathbb{E}[\mathcal{L}(\alpha_n^{\mathcal{L}}) + \mathcal{L}(\alpha_{n,M}^{\mathcal{L}})] \\
 &\leq \frac{1}{\rho_{\min}} \mathbb{E}[\mathcal{L}(\alpha_n^{\mathcal{L}}) + \mathcal{L}^M(\alpha_{n,M}^{\mathcal{L}})] + \frac{1}{\rho_{\min}} \mathbb{E}[\mathcal{L}(\alpha_{n,M}^{\mathcal{L}}) - \mathcal{L}^M(\alpha_{n,M}^{\mathcal{L}})] \\
 &\leq \frac{1}{\rho_{\min}} \mathbb{E}[\mathcal{L}(\alpha_n^{\mathcal{L}}) + \mathcal{L}^M(\alpha_n^{\mathcal{L}})] + \frac{2}{\rho_{\min}} \mathcal{R}_M(\mathcal{F}_n^{\mathcal{L}}) \\
 &\leq \frac{1}{\rho_{\min}} \mathbb{E}[\mathcal{L}^M(\alpha_n^{\mathcal{L}}) - \mathcal{L}(\alpha_n^{\mathcal{L}})] + \frac{2}{\rho_{\min}} \mathbb{E}[\mathcal{L}(\alpha_n^{\mathcal{L}})] + \frac{2}{\rho_{\min}} \mathcal{R}_M(\mathcal{F}_n^{\mathcal{L}}) \\
 &\leq \frac{4}{\rho_{\min}} \mathcal{R}_M(\mathcal{F}_n^{\mathcal{L}}) + 2\kappa(A) \inf_{\alpha \in \mathcal{N}_n} \|\alpha - \alpha^*\|_{L^1(\Omega;\ell^2)}. \tag{11}
 \end{aligned}$$



2. Final result

Final result: If for all $n \in \mathbb{N}$, $\mathcal{R}_M(\mathcal{F}_n^{\mathcal{L}}) \rightarrow 0$ as $M \rightarrow \infty$, then

$$\mathbb{E}[\|u_h - u_{h,n,M}\|_{L^1(\Omega; L^2(D))}] \leq C \left(\frac{1}{\rho_{\min}} \mathcal{R}_M(\mathcal{F}_n^{\mathcal{L}}) + \kappa(A) \inf_{\alpha \in \mathcal{N}_n} \|\alpha - \alpha^*\|_{L^1(\Omega; \ell^2)} \right) \rightarrow 0$$

as $n, M \rightarrow \infty$. (12)

Explicit convergence rate: For nonsymmetric case, no direct estimate for $\frac{1}{\rho_{\min}}$.



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3. Explicit error for self-adjoint PDEs

Self-adjoint case: A is symmetric and positive definite, i.e., $\rho_{\min} = \lambda_{\min}(A) \geq Ch^d$,

$$\rho_{\max} = \lambda_{\max} \leq Ch^{d-2}.$$

$$\kappa(A) = \frac{\rho_{\max}}{\rho_{\min}} = \frac{\lambda_{\max}(A)}{\lambda_{\min}(A)} \leq Ch^{-2}. \quad (13)$$

Still need to estimate $\mathcal{R}_M(\mathcal{F}_n^{\mathcal{L}})$ and $\inf_{\alpha \in \mathcal{N}_n} \|\alpha - \alpha^*\|_{L^1(\Omega; \ell^2)}$ to obtain the explicit convergence rate.



3. Estimate of $\inf_{\alpha \in \mathcal{N}_n} \|\alpha - \alpha^*\|_{L^1(\Omega; \ell^2)}$

Theorem (Barron (1993); E et al. (2022))

Let $g \in \mathcal{W}(\Omega)$ and $n \in \mathbb{N}$. Then there exists a two-layer ReLU neural network $g_\theta \in \mathcal{N}_n$ with the parameter $\theta = (a_j, b_j, c_j)_{j=1}^n$ and $\mathcal{N}_n$ collecting $\frac{1}{n} \sum_{j=1}^n a_j \sigma(b_j \cdot \omega + c_j)_{i=1}^n$ such that $\|\theta\|_{\mathcal{P}} \leq 2\|g\|_{\mathcal{B}^2(\Omega)}$, and

$$\|g_\theta - g\|_{L^2(\Omega; \ell^2)} \leq C \|g\|_{\mathcal{B}^2(\Omega; \ell^2)} n^{-\frac{1}{2}}. \quad (14)$$

$$\|\theta\|_{\mathcal{P}} := \left\| \frac{1}{n} \sum_{j=1}^n a_j (\|b_j\|_{\ell^1} + |c_j|) \right\|_{\ell^\infty}.$$

$$\|g\|_{\mathcal{B}^s(\Omega)} := \inf_{g_e|_{\Omega=g}} \int_{\mathbb{R}^m} (1 + |\xi|)^s |\hat{g}_e(\xi)| d\xi, \hat{g}_e \text{ is the Fourier transform of } g_e$$

$$\inf_{\alpha \in \mathcal{N}_n} \|\alpha - \alpha^*\|_{L^1(\Omega; \ell^2)} \leq C \inf_{\alpha \in \mathcal{N}_n} \|\alpha - \alpha^*\|_{L^2(\Omega; \ell^2)} \leq C n^{-\frac{1}{2}} \|\alpha^*\|_{\mathcal{B}^2(\Omega; \ell^2)}. \quad (15)$$



3. Estimate of $\|\alpha^*\|_{\mathcal{B}^2(\Omega)}$

Since $\alpha^*(\omega) = A^{-1}L(\omega)$ and $L = \int_D f(\mathbf{x}, \omega) \Phi(\mathbf{x}) d\mathbf{x}$ with $\Phi = (\phi_1, \phi_2, \dots, \phi_{N_h})^\top$, we have

$$\hat{\alpha}^*(\xi) = A^{-1} \int_D \hat{f}(\mathbf{x}, \xi) \Phi(\mathbf{x}) d\mathbf{x}. \quad (16)$$

Then we have

$$\begin{aligned} \|\alpha^*\|_{\mathcal{B}^2(\Omega; \ell^2)} &= \int_{\mathbb{R}^m} (1 + |\xi|)^2 \|\hat{\alpha}^*(\xi)\|_{\ell^2} d\xi \\ &\leq \int_{\mathbb{R}^m} (1 + |\xi|)^2 \int_D |\hat{f}(\mathbf{x}, \xi)| \|A^{-1} \Phi(\mathbf{x})\|_{\ell^2} d\mathbf{x} d\xi. \end{aligned} \quad (17)$$

Since $\|A^{-1} \Phi\|_{\ell^2} \leq \frac{1}{\rho_{\min}} \|\Phi(\mathbf{x})\|_{\ell^2} \leq \frac{N_h^{1/2}}{\rho_{\min}} \sim h^{-3d/2}$, we have

$$\|\alpha^*\|_{\mathcal{B}^2(\Omega; \ell^2)} \leq Ch^{-3d/2} \int_D \int_{\mathbb{R}^m} (1 + |\xi|)^2 |\hat{f}(\mathbf{x}, \xi)| d\mathbf{x} d\xi \sim h^{-3d/2} \|f\|_{L^1(D; \mathcal{B}^2(\Omega))}. \quad (18)$$



3. Estimate of $\mathcal{R}_M(\mathcal{F}_n^{\mathcal{L}})$

Theorem (Neyshabur et al., 2015; E et al., 2022)

For the function class $\mathcal{N}_n^* = \{g_\theta \in \mathcal{N}_n : \|\theta\|_{\mathcal{D}} \leq Q\}$, we have

$$\mathcal{R}_M(\mathcal{N}_n^*) \leq c \frac{QN_h}{\sqrt{M}}. \quad (19)$$

$$\begin{aligned} \mathcal{R}_M(\mathcal{F}_n^{\mathcal{L}}) &= \mathbb{E}_{\{X_i\}_{i=1}^M} \mathbb{E}_{\{\varepsilon\}_{i=1}^M} \left[\sup_{\alpha \in \mathcal{N}_n^*} \frac{1}{M} \sum_{i=1}^M \varepsilon_i \|A\alpha(\omega^i) - L\|_{\ell^2} \right] \\ &\leq \mathbb{E}_{\{\omega^i\}_{i=1}^M} \mathbb{E}_{\{\varepsilon\}_{i=1}^M} \left[\sup_{\alpha \in \mathcal{N}_n^*} \frac{1}{M} \sum_{i=1}^M \varepsilon_i \|A\alpha(\omega^i)\|_{\ell^2} \right] \\ &\leq \rho_{\max} \mathbb{E}_{\{\omega^i\}_{i=1}^M} \mathbb{E}_{\{\varepsilon\}_{i=1}^M} \left[\sup_{\alpha \in \mathcal{N}_n^*} \frac{1}{M} \sum_{i=1}^M \varepsilon_i \|\alpha(\omega^i)\|_{\ell^2} \right] = \rho_{\max} \mathcal{R}_M(\mathcal{N}_n^*) \leq c \frac{Q}{\sqrt{M}} h^{-2}. \end{aligned} \quad (20)$$



3. Final convergence rate

Final convergence rate: If $\|f\|_{L^1(D; \mathcal{B}^2(\Omega))} < \infty$, then

$$\begin{aligned}\mathbb{E}\left[\|u - u_{h,n,M}\|_{L^1(\Omega; L^2(D))}\right] &\leq \mathbb{E}\left[\|u - u_h\|_{L^1(\Omega; L^2(D))}\right] + Cn^{-\frac{1}{2}}h^{-\frac{3d}{2}} + CQh^{-2}M^{-\frac{1}{2}} \\ &\leq Ch^{\ell+1} + Cn^{-\frac{1}{2}}h^{-\frac{3d}{2}} + CQh^{-2}M^{-\frac{1}{2}} \\ &= Ch^{\ell+1} + Cn^{-\frac{1}{2}}\kappa(A)^{\frac{3d}{4}} + CQ\kappa(A)M^{-\frac{1}{2}}.\end{aligned}\tag{21}$$



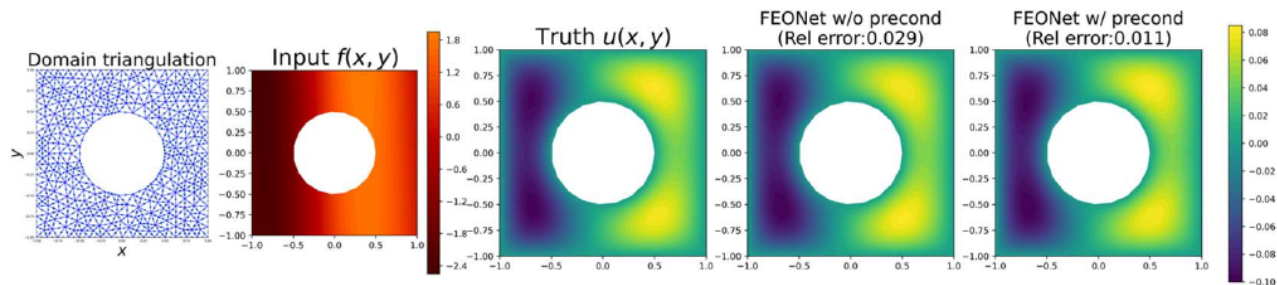
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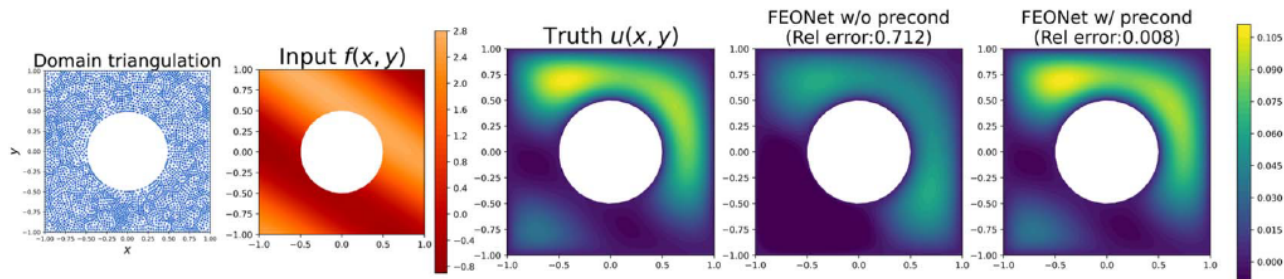
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4. Numerical Experiments



(a) Results for Poisson equation using P2 on 551 elements



(b) Results for Poisson equation using P2 on 1530 elements



4. Numerical Experiments

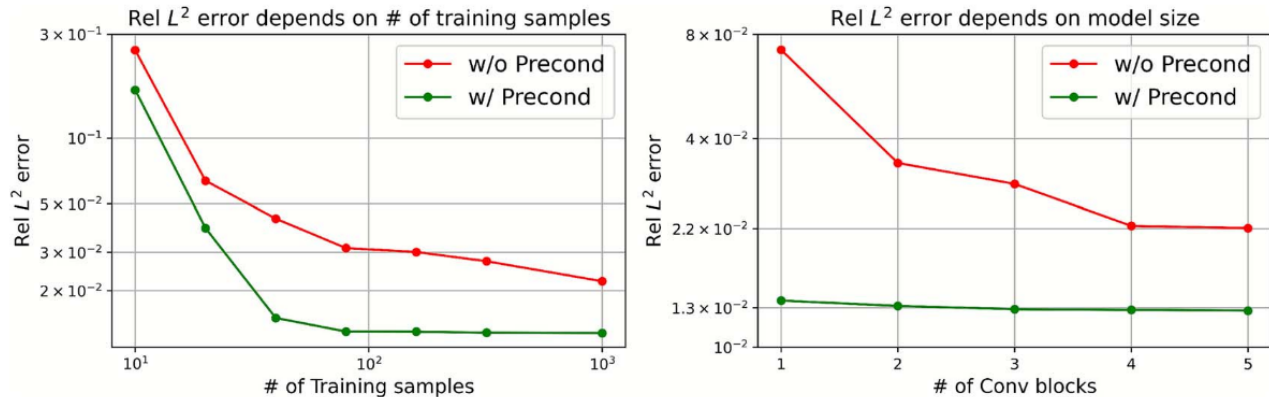


FIG. 5. The relative L^2 errors resulting from varying the number of training samples and the model size. For interpretation of the references to color in this figure, the reader is referred to the online version of this article.



4. Numerical Experiments

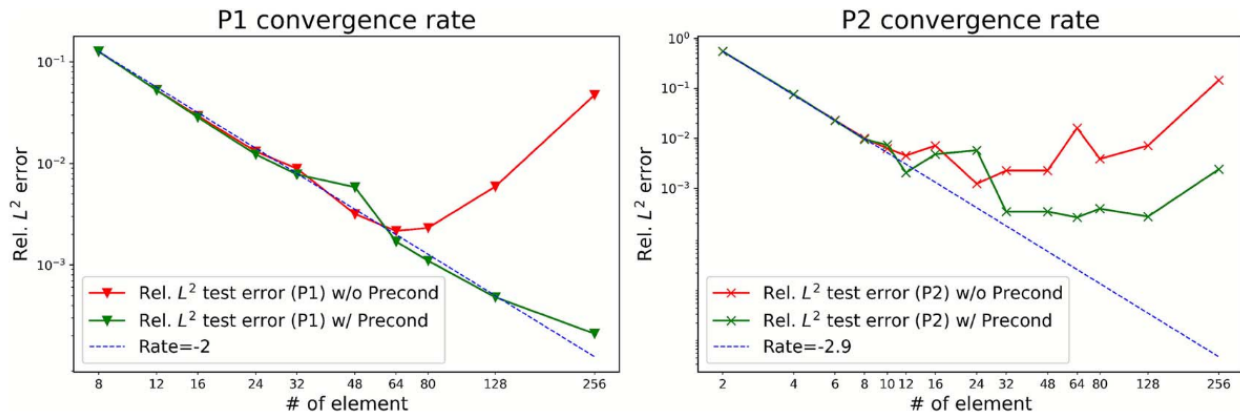


FIG. 7. The relative L^2 errors against the number of elements for the convection–diffusion equation with and without preconditioning. For interpretation of the references to color in this figure, the reader is referred to the online version of this article.



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5. Conclusion

Equation: $-\operatorname{div}(a(x)\nabla u) + b(x) \cdot \nabla u + c(x)u = f(x) \quad \text{in } D, \quad u|_{\partial D} = g(x).$

Solution Structure: $u = \sum_{i=1}^{N_h} (\hat{\alpha}_{n,M}^{\mathcal{L}})_i \phi_i(t, x)$

Loss function: $\mathcal{L}^M(\alpha) = \frac{|\Omega|}{M} \sum_{j=1}^M \|A\alpha(\omega^j) - L\|_{\ell^2}$

Convergence rate: $Ch^{\ell+1} + Cn^{-\frac{1}{2}}\kappa(A)^{\frac{3d}{4}} + CQ_{\kappa}(A)M^{-\frac{1}{2}}$

Limitations:

1. Obtain the explicit convergence rate for nonsymmetric case
2. Obtain the convergence rate for more general input, e.g., $\{f(x_i)\}_{i=1}^N$

Future work: nonlinear PDEs, time-dependent PDEs, complex geometries, etc.



The Parametric Complexity of Operator Learning

Thank you for listening!

Any questions?